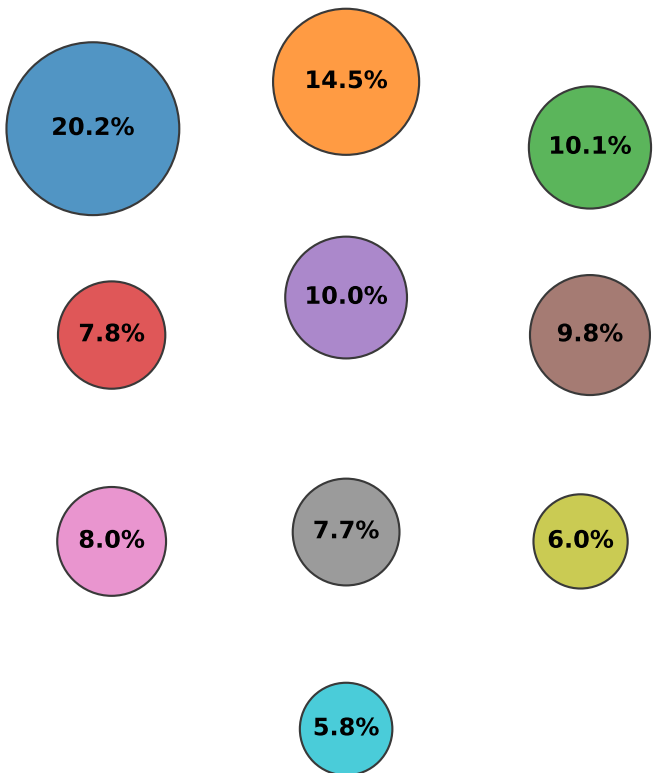
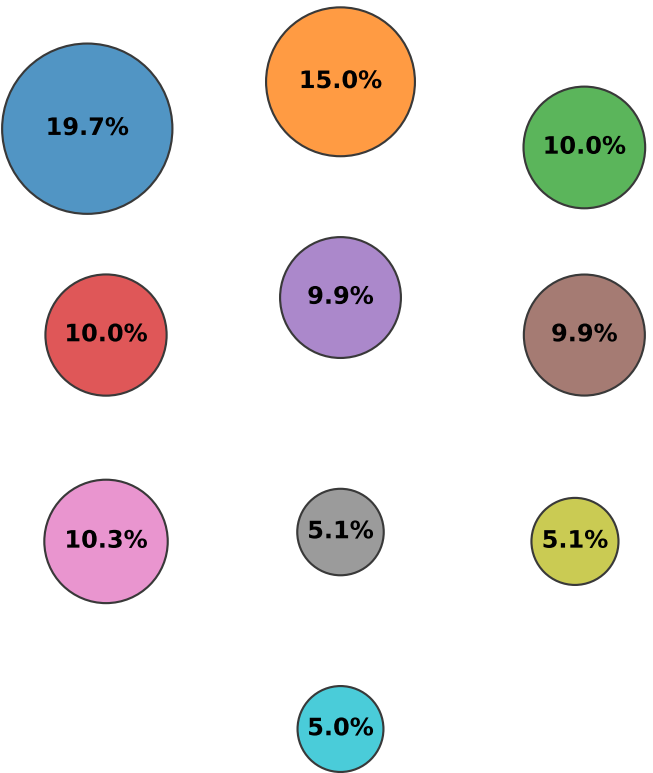


Three Perspectives on the Thematic Landscape of ML-OPF Research

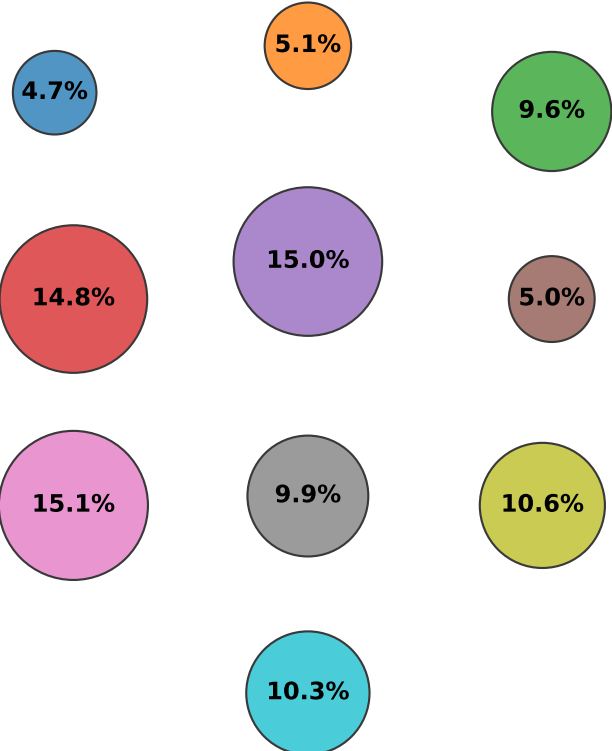
Academic Research (Scopus)



Patent-Cited Research (Lens)



Policy-Cited Research (Overton)



- Metaheuristics & computational optimization
- ML, forecasting & predictive analytics
- OPF & mathematical optimization
- State estimation, monitoring & cybersecurity
- Reliability, resilience & stability
- Distributed control & grid operation
- Decarbonization, electrification & integrated energy
- Energy management, markets & flexibility
- Smart grids, IoT & digitalization
- Renewables, storage & uncertainty